Part C in one page — market making on hftbacktest: what made money, what didn't

The question behind all of Part C: can reinforcement learning improve a published market-making strategy, measured on days it never saw? This page is the summary; each phase has its own writeup, linked in §4.

> **At a glance**
>
> - **Best held-out result:** OBI on XRPUSDC with a markout-driven spread, **+4.91%** over eleven held-out sessions, **+1.38% without the crash day**, positive on **9 of 11** sessions at ~500 fills a day.
> - **Most robust held-out result:** OBI on SUIUSDC (C8, seven held-out sessions on a 24/7 split), **+2.52%** at 175 fills a day, and still positive without the crash (+1.77%), without its best session (+1.52%), without the maker rebate (+2.05%), with a 1 bp maker fee (+1.13%) and at 4× order latency (+1.30%). XRP's OBI gain, by the same stress tests, is the crash session.
> - **The fair price matters more than anything else tried — if it were known.** Quoting around the mid 1–3 s ahead (C9) turns ETH GLFT from −3.25% to +4.30–19.67% held out; but the best forecast of the mid explains 2–10% of its 1 s move, where about 40% would be needed, and no forecast, large-move rule or DAgger student captures the gain.
> - **Everything on ETHUSDC still loses held out**, bar the rebate. The best, the same rule on OBI, is at −0.62%, against −6.40% for the widened baseline and −7.14% for the published parameters.
> - **The two things that helped were both hand-written rules:** a wider, per-asset spread floor (C6) and widening after toxic fills (C7).
> - **Learning did not beat them.** TD3 on C6's widened strategies (two recipes, five seeds), GRPO- and PPO-style parameter search, extra order-book and BTCUSDT inputs, fill targets, a reward that discounts inventory PnL and networks up to 170 times larger all lose to the best fixed rule held out; so does RL on top of that rule (C8). A circuit breaker helps the losing strategies on ETH; extra controls chosen on training data hurt OBI.

1. Setup, in five lines

- **Data:** Binance USDⓈ-M perpetual L2 books and trades, 2025-08-26 to 2025-09-25, replayed tick by tick in hftbacktest with queue position and synthesised order latency.
- **Split:** everything is chosen on sessions before 2025-09-15 and reported on the eleven sessions from 09-15 to 09-25 (C1–C7, C4). C8 re-splits the same month into 24 training and 7 held-out sessions (09-19 to 09-25). Nothing is chosen on held-out data.

- **Deep assets:** ETHUSDC and XRPUSDC, 11 held-out sessions each. Seven more assets with 3 held-out sessions each.
- **Strategies:** the five market makers hftbacktest publishes — GLFT, a volatility grid (`hfgrid`), order-book imbalance (`obi`), and two pair strategies (`apt`, `basis`) on XRP.
- **Score:** cumulative held-out return in % of a 100,000 book; fills and per-session consistency beside it, so "stopped trading" never passes for "made money".

2. The track record

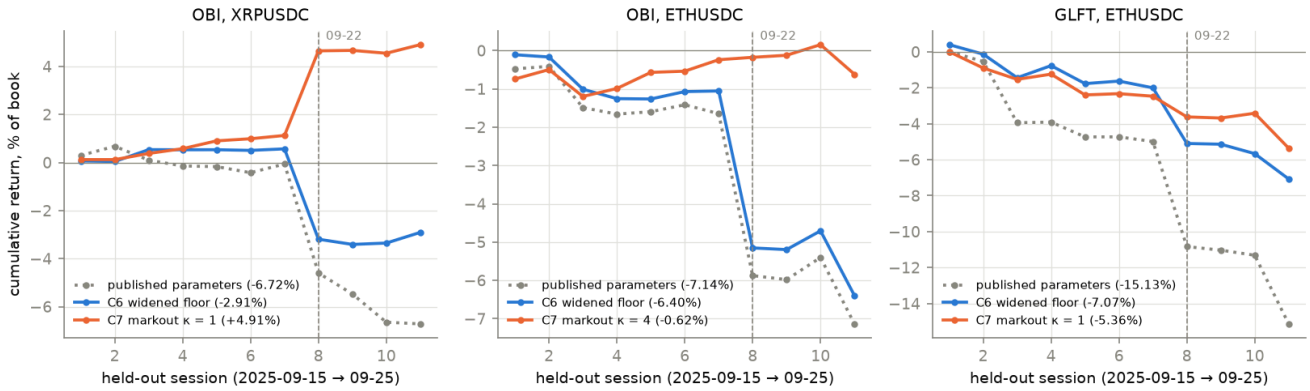


Figure — cumulative return over the eleven held-out sessions: the published parameters (dotted), C6's widened spread floor (blue), and C7's markout rule on top of it (orange). Session 8 is 2025-09-22, when the market crashed.

Two things are visible at once. The widened floor helps on every panel but does not stop the crash-day loss. The markout rule does: on 09-22 the widened strategies lose 3–4% while the markout versions lose about 1% or gain — plausibly because they had already widened as their fills turned toxic. Part of XRP's gain that day is holding the right inventory through the move, which is why the result is also quoted with 09-22 removed.

3. Every method, held out

asset	strategy	published	C6 widened	C6 RL v2	C7 markout	C4 grid (20 d)	C4 GRPO
ETH	glft	−15.13	−7.07	−6.30	−5.36	−3.01	**−1.89**
ETH	hfgrid	−1.98	−2.11	−2.20	−1.99	**−1.86**	–
ETH	obi	−7.14	−6.40	−8.00	**−0.62**	**−0.62**	−2.91
XRP	glft	−13.17	−1.89	−0.86	**−0.55**	−1.22	–
XRP	hfgrid	−1.58	−1.15	−1.48	**+1.96**	+0.59 †	–
XRP	obi	−6.72	−2.91	−2.15	**+4.91**	+3.64	–
XRP	apt	−3.39	+2.00	−1.66	–	+7.26 †	–
XRP	basis	−4.59	+0.36	−2.35	–	+1.69 †	–

Held-out cumulative return, % of book, eleven sessions. RL and GRPO are medians over seeds. † trades a handful of fills or none on a median session; the number is one crash session, not market making.

3.1 On C8's split, with stress tests

C8 moved to 24 training sessions and kept the last seven held out, then stress-tested everything that made money: maker fee without the rebate, higher latency, pessimistic queue models, neighbouring parameters, and without the crash or the best session.

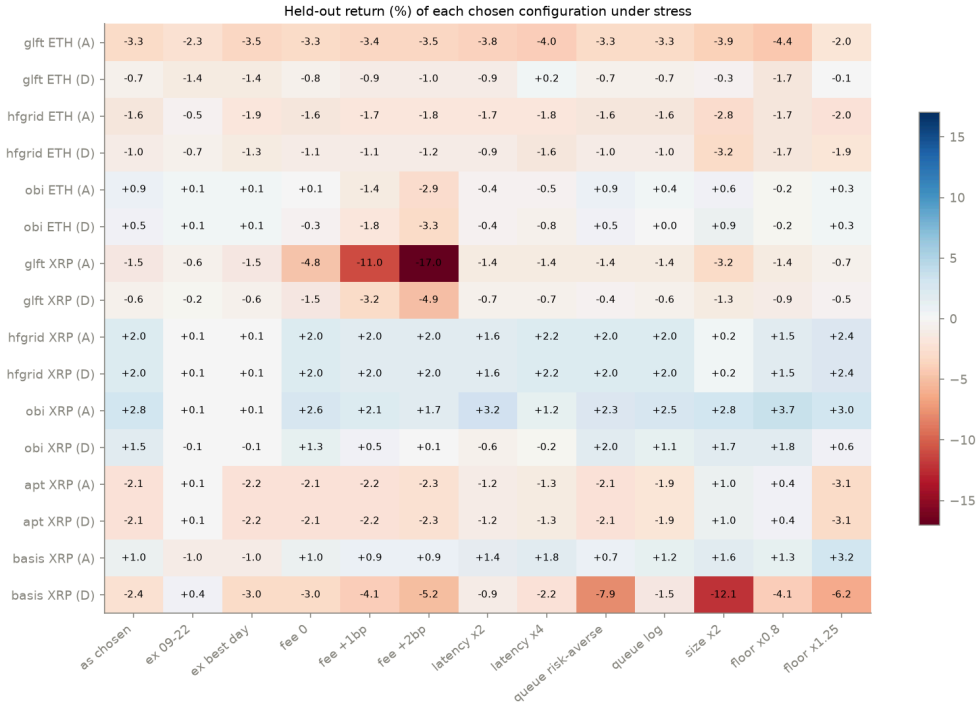


Figure — from C8 §6: held-out return of each chosen configuration under each stress.

configuration	held out	ex 09-22	ex best day	fee 0	fee +2 bp	latency ×4	fills/day	verdict
SUIUSDC OBI (floor 8 bp, κ 4)	**+2.52**	**+1.77**	**+1.52**	**+2.05**	+0.21	**+1.30**	175	makes money on ordinary sessions, robust to costs; sensitive to κ
XRPUSDC OBI (floor 12 bp, κ 2, skew ×2)	+2.84	+0.09	+0.09	+2.61	+1.68	+1.17	106	robust to costs; the gain is the crash
XRPUSDT OBI (floor 8 bp, κ 2)	+3.53	+0.48	+0.48	+3.17	+1.74	+2.84	189	as XRPUSDC
AVAXUSDC OBI (floor 16 bp, κ 0)	+2.68	+2.34	−0.37	+2.35	+1.04	+3.14	150	one session
ETHUSDC OBI (floor 6 bp, κ 4)	+0.85	+0.14	+0.14	+0.09	−2.92	−0.48	369	the rebate

configuration	held out	ex 09-22	ex best day	fee 0	fee +2 bp	latency ×4	fills/day	verdict
BNBUSDC GLFT (floor 2 bp, κ 1)	+1.72	+1.53	+1.23	+0.75	−3.05	+1.71	489	the rebate at higher fees

Held-out cumulative return, % of book, seven sessions (09-19 to 09-25).

4. What each phase contributed

phase	question	answer, held out
C1	Does calibrated GLFT make money here?	No: it quotes inside the cost of adverse selection and loses 0.8 bp per fill.
C2	Can the problem be posed for RL without a degenerate answer?	Yes, once the "quote so far out nothing fills" corner is ruled out.
C3	Do DDPG/TD3/SAC beat the fixed formula?	They lose less than it, never make money; seed spread is large.
C6	Is that just GLFT being a bad baseline?	No. On five strategies and nine assets, a per-asset wider spread is the largest single gain (+2.25 points median); TD3 on top of it does not beat it.
C7	What information is the policy missing?	Not the deep book or BTCUSDT. The toxicity of its own recent fills: a markout rule beats C6 in 6 of 6 pairs and 13 of 18 untuned.
C4	Can search (grid, GRPO, PPO) tune the fixed rules further?	The grid refines them on ETH; learned search, fill targets, an inventory-discounted reward and larger networks do not.
C9	Can the fair price be forecast ~1 s ahead, and does denoising help anywhere?	Foresight of 1–3 s would be worth up to +40 points held out (little on XRP); real forecasts reach R^2 0.02–0.10 and help one asset (AVAX). DAgger with a future-seeing teacher halves ETH GLFT's loss but does not make it pay. No denoiser helps on any of nine assets.
C8	Do risk controls, denoising or RL on top of the markout rule help, and does what makes money survive stress?	A circuit breaker helps the losers on ETH; no denoiser, combination of controls or RL policy improves OBI. SUIUSDC OBI survives every cost, latency and session stress but κ ×0.5; XRP's gain is the crash, ETH's the rebate.

5. Lessons that held up

1. **Measure the cost of a fill before tuning anything.** Every positive result in Part C comes from quoting wider when fills are expensive.
2. **Fixed rules first.** Each learned method was beaten by a hand-written rule that encoded the right quantity; RL is worth trying on top of the best rule, not instead of it.
3. **Seeds and sessions, not single runs.** Single-seed RL rows were systematically kinder than five-seed rows, and four training sessions chose XRP parameters that twenty did not.
4. **Report the crash day separately.** One session (2025-09-22) can make a strategy that barely trades look like the best in the table.

5. **Fills are not free.** Both forcing volume and rewarding it cost money on this venue, at every level tried.
6. **Stress-test before believing a gain.** Of six configurations that made money held out in C8, one survives removing the crash, the best session and the rebate; the others were one session or the rebate.
7. **More knobs fit the training month.** Adding controls chosen on 24 sessions helped strategies that lose and hurt the one that works.
8. **Measure the ceiling before learning.** C9's oracle showed where the money is (the next 1–3 s) and its noisy oracles how good a forecast must be ($R^2 \approx 0.4$); real forecasts at 0.02–0.10 were then no surprise.

6. What next

C8 did the follow-ups listed here before it (circuit breaker, loss limits, stress tests, breadth assets on more held-out sessions, RL on top of the markout rule). What remains:

- **The forward test:** the frozen SUIUSDC and XRP OBI choices on sessions after 2025-09-25, with no re-selection. Seven held-out sessions cannot tell a +2.5% edge from luck.
- **OBI on more assets**, since it is the only strategy that generalises.
- **A crash-aware breaker** that does not pause when the position is already on the right side of the move — the breaker's cost on OBI XRP is exactly the crash session.
- **κ from the asset's markout**, since SUIUSDC OBI depends on $\kappa = 4$ and only the grid chooses κ today.
- **Information from outside the book** for the rare large moves C9 found to carry the value: other venues' perpetuals, spot, funding and liquidation feeds (C9 §12).